treat our data structure as a black box: we focus on how to use it, ignoring the details of its implementation.

Only in the next section we start discussing how a data structure works. We focus on describing the mechanism, using figures and pseudo-code examples to clarify how it works.

After the code section, we are going to discuss advanced theory topics like performance or mathematical proofs for the algorithms.

Usually we also provide a list of additional applications that make use of the algorithms presented, although for most of these further examples we have to omit coding, for the sake of space.

2.2 The problem: handling priority

The first problem we are going to tackle is handling tasks based on priority. This is something all of us are familiar with, in some way.

The problem can be described in these terms: given a collection of tasks with different priorities, determine which task should be executed next.

We can find many examples in the real world where we apply, consciously or not, techniques that help us decide what to do next. Our daily lives are full of tasks: usually the